Department of Electrical Engineering
University of Arkansas

ELEG 3143 Probability & Stochastic Process
Ch. 2 Random Variables

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OUTLINE

- Random Variable
- Discrete Random Variable
- Continuous Random Variable
- Moments of Discrete RV
- Moments of Continuous RV
- Functions of one RV

RANDOM VARIABLE

- **Random variable (RV)**
  - A function, which assigns a unique numerical value to each possible outcome of a random experiment
  - A mapping from the sample space to some real numbers.
  - Example
    - Toss a coin. S = {H, T}
    - Define random variable X such that
      - X = 1, if the outcome is H
      - X = 0, if the outcome is T
    - A mapping from the sample space to some real numbers
      - Event {H} → X = 1
      - Event {T} → X = 0
    - The value of the random variable, X, is random. The actual value of X depends on the outcome of the experiment
    - We can assign probability to each value of the random variable
      - P(X = 1) = P({H}) = 0.5
      - P(X = 0) = P({T}) = 0.5

RANDOM VARIABLE

- **Example**
  - Consider an experiment of tossing 2 fair coins. Define the RV, Y, as the number of heads appearing.
    - Find the mapping from the sample space to the value of Y
    - Find the probability of the different values of Y

- Capital letter is used to represent an RV
- Small letter is used to represent a deterministic RV.

RANDOM VARIABLE

- **Example**
  - Suppose that there is an unfair coin with a probability *p* of coming up heads. We perform an experiment, in which we flip the coin, until the first head appears. Define an RV, N, as the number of flips required.
    - 1. Find the mapping from the sample space to the RV
    - 2. Find the probability of the RV.

RANDOM VARIABLE

- **Discrete RV v.s. continuous RV**
  - Consider a random variable X
  - If X has countable number values, then X is a discrete RV
    - E.g. toss a coin, X = 0 if H, X = 1 if T
    - E.g. Defined the RV N as the number of flips required to get the head of a coin
  - If X has uncountable number of values, then X is a continuous RV
    - E.g. Define the RV X as the current temperature
- **Example**
  - Consider the lifetime of a car
    - Define the RV, X, as the lifetime of the car
      - Then X is a continuous RV, $X \in [0,+\infty)$
    - Define the RV, Y, as follows
      - If the car has a life longer than 2 years → Y = 1
      - If the car has a life shorter than 2 years → Y = 0
      - Then Y is a discrete RV, $Y \in \{0,1\}$

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DISCRETE RV

- **Discrete RV**
  - A random variable that can take at most a countable number of possible values.
- **Probability mass function (PMF)**
  - Consider a discrete RV X, which can take any one of the following values: $a_1, a_2, a_3, \cdots, a_n, \cdots$
  - The probability mass function of X is defined as

$$p_X(a_i) = \Pr(X = a_i) \qquad i = 1,2,3,\cdots$$

  - Example
    - Flip an unfair coin with the probability p of coming up head. Define an RV X, such that H → X = 1; T → X = 0. What is the PMF?

DISCRETE RV

- **Example**
  - Consider an urn with 3 black balls, 5 red balls, and 8 white balls. Randomly pick 1 ball. Define an RV X as:
    - black ball → X = 1, red ball → X = 2, white ball → X = 3
    - Find the PMF.

DISCRETE RV

- **PMF**
  - For an RV $X \in \{a_1, a_2, \cdots, a_n, \cdots\}$, since X must take one of the values of a_n

$$\sum_{i=1}^{\infty} p_X(a_i) = \sum_{i=1}^{\infty} \Pr(X = a_i) = 1$$

- **Cumulative distribution function (CDF)**
  - All called distribution function

$$F_X(x) = \Pr(X \le x)$$

  - The probability that the RV, *X*, is less than or equal to a deterministic number, *x*
    - capital letter X: Random variable (has random values)
    - Small letter x: deterministic variable (has fixed value)
    - E.g. $F_X(3) = \Pr(X \le 3)$

DISCRETE RV

- **Example**
  - Flip an unfair coin with the probability *p* of coming up head. Define an RV X, such that H → X = 1; T → X = 0. What is the CDF?

DISCRETE RV

- **Example**
  - Suppose X has a PMF given by

$$p_X(1) = \frac{1}{2} \qquad p_X(2) = \frac{1}{3} \qquad p_X(3) = \frac{1}{6}$$

    - Find and plot the CDF.

DISCRETE RV

- **Properties of CDF**
  - $0 \le F_X(x) \le 1$
  - $F_X(-\infty) = 0$
  - $F_X(\infty) = 1$
  - $F_X(x)$ is a non-decreasing function of x
  - $\Pr(x_1 < X \le x_2) = F_X(x_2) - F_X(x_1)$
  - $\Pr(X > x) = 1 - F_X(x)$

DISCRETE RV

- **Bernoulli RV**
  - Consider an experiment, whose outcome can be classified as either a "success" or as a "failure". The probability of success is *p*. Define an RV X such that
    - Success → X = 1
    - Failure → X = 0.

    Then X is called a Bernoulli RV
  - The PMF

$$\Pr(X = 1) = p \qquad \Pr(X = 0) = 1 - p$$

  - The CDF

  - Recall Bernoulli trials
    - Perform the experiment independently for *N* times

DISCRETE RV

- **Binomial RV**
  - Perform *n* independent trials, each of which results in a "success" with probability *p*. Define the RV Y as the total number of successes that occur in the *n* trials. Then Y is a binomial RV.
  - PMF

$$p_Y(k) = \Pr(Y = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

  - The summation of PMF

$$\sum_{k=0}^{n} p_Y(k) =$$

DISCRETE RV

- **Relationship between Bernoulli RV and Binomial RV**
  - Consider an experiment, whose outcome can be classified as either a "success" or as a "failure". The probability of success is *p*. The Bernoulli RV, X, is defined as
    - Success → X = 1
    - Failure → X = 0.

$$X \sim \text{Bernoulli}\ (p)$$

  - Perform *n* independent trials, each of which results in a "success" with probability *p*. Define the RV Y as the total number of successes that occur in the *n* trials. Then Y is a binomial RV. X_i
    - Each trial results in a Bernoulli RV

$$X_i \sim \text{Bernoulli}\ (p) \qquad Y \sim \text{Binomial}\ (n, p)$$

  - The relationship between the Bernoulli RV X_i, and the Binomial RV Y

$$Y = \sum_{i=1}^{n} X_i$$

DISCRETE RV

- **Example**
  - It is known that any item produced by a certain machine will be defective with probability 0.1, independently of any other item. What is the probability that in a sample of 3 items, at most one will be defective?

DISCRETE RV

- **Binomial RV**
  - On a multiple-choice exam with 4 possible answers for each of the 5 questions. What is the probability that a student would get 4 or more correct answers just by guessing?

DISCRETE RV

- **Geometric RV**
  - Suppose that independent trials, each having probability p of being success. The experiment is performed until a success occurs. Define X as the number of trials required until the first success. Then X is said to be a geometric RV.
  - PMF

$$p_X(k) = \Pr(X = k) = (1-p)^{k-1} p$$

  - The sum of Geometric PMF

$$\sum_{k=1}^{\infty} \Pr(X = k) =$$

DISCRETE RV

- **Poisson RV**
  - A random variable X, taking on the values 0, 1, 2, ……, is said to be a Poisson random variable with parameter $\lambda > 0$, if the PMF can be written as

$$P(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}$$

- **Sum of Poisson PMF**

$$\sum_{k=0}^{+\infty} e^{-\lambda} \frac{\lambda^k}{k!} =$$

$$e^{\lambda} = \sum_{k=0}^{\infty} \frac{\lambda^k}{k!}$$

DISCRETE RV

- **Example**
  - If the number of accidents occurring on a highway each day is a Poisson RV with parameter $\lambda = 0.1$. What is the probability that at least 1 accident occur today?

DISCRETE RV: SUMMARY

- **Bernoulli RV**

$$P(X=1)=p \qquad P(X=0)=1-p$$

- **Binomial RV**

$$P(X=k)=\binom{n}{k}p^k(1-p)^{n-k}$$

- **Geometric RV**

$$P(X=k)=(1-p)^{k-1}p$$

- **Poisson RV**

$$P(X=k)=e^{-\lambda}\frac{\lambda^k}{k!}$$

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CONTINUOUS RV

- **Continuous RV**
  - If a random variable can take an unaccountable number of values, then the random variable is a continuous random variable.
  - Examples of continuous RV
    - The daily average temperature
    - The expected lifetime of a computer
    - The amplitude of noise in an electronic component
    - ……

CONTINUOUS RV

- **Cumulative distribution function (distribution function)**
  - The cumulative distribution function (CDF) of a continuous RV is

$$F_X(x) = \Pr(X \le x)$$

    - The probability that the RV X is smaller than or equal to x
  - Recall: the CDF of a discrete RV Y is $F_Y(y) = \Pr(Y \le y)$
  - Properties of CDF

$$0 \le F_X(x) \le 1$$

$$F_X(-\infty) = 0$$

$$F_X(\infty) = 1$$

$$\Pr(x_1 < X \le x_2) = F_X(x_2) - F_X(x_1)$$

$F_X(x)$ is a non-decreasing function of x

CONTINUOUS RV

- **Example**
  - A particular random variable has a probability distribution function given by

$$F_X(x) = \begin{cases} 1 - e^{-2x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

    - What is the probability that $X \leq 0.25$
    - What is the probability that $X > 0.5$
    - What is the probability that $0.25 < X \leq 0.5$

CONTINUOUS RV

- **Probability density function (pdf)**

$$f(x) = \frac{dF(x)}{dx}$$

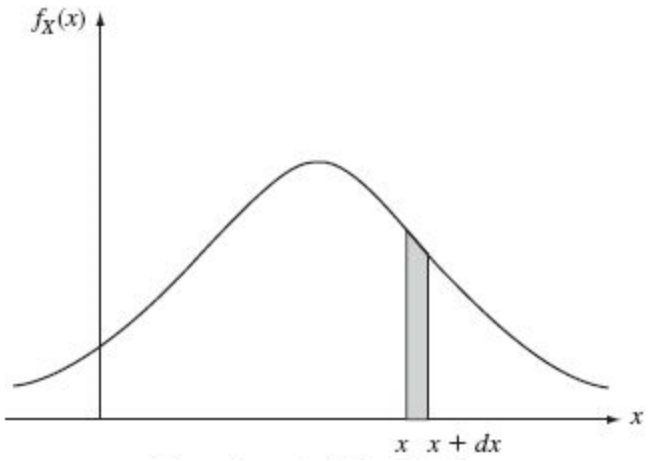
  - The "density" of probability
  - Review: differentiation

$$\frac{dF(x)}{dx} = \lim_{\Delta \to 0} \frac{F(x+\Delta) - F(x)}{\Delta}$$

  - Interpretation:

$$f(x) \approx \frac{P(x < X \le x + \Delta)}{\Delta}$$

$$P\{x < X \le x + \Delta\} = f(x)\Delta$$



CONTINUOUS RV

- **Relationship between pdf and CDF**

$$f_X(x) = \frac{dF_X(x)}{dx} \qquad F_X(x) = \int_{-\infty}^{x} f_X(y)dy$$

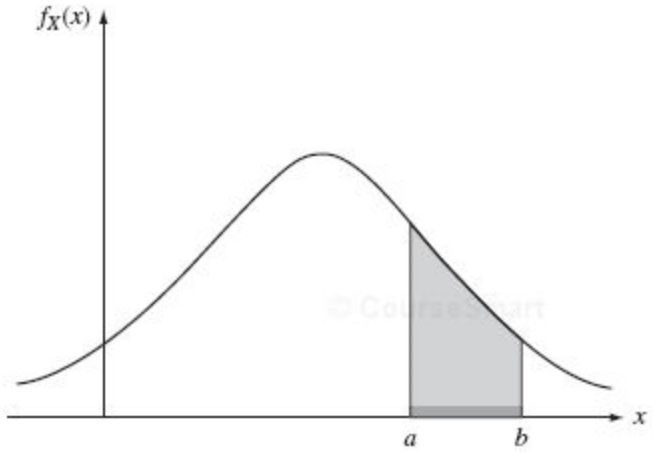
– $\Pr(X \le a) = F_X(a) = \int_{-\infty}^{a} f_X(y)dy$

$$\Pr(X \le b) = F_X(b) = \int_{-\infty}^{b} f_X(y)dy$$

$$\Pr(a < X \le b) = F_X(b) - F_X(a) = \int_{a}^{b} f_X(y)dy$$

– Review: integration
  - finding the area under the integrand

$$\int_a^b f_X(y)dy \approx \sum f_X(y_i)\Delta$$



CONTINUOUS RV

- **Properties of pdf**

$$f(x) \geq 0$$

$$P\{a < X \leq b\} = \int_a^b f(x)dx$$

$$\int_{-\infty}^{b} f(x)dx = F(b)$$

$$\int_{-\infty}^{+\infty} f(x)dx = 1$$

CONTINUOUS RV

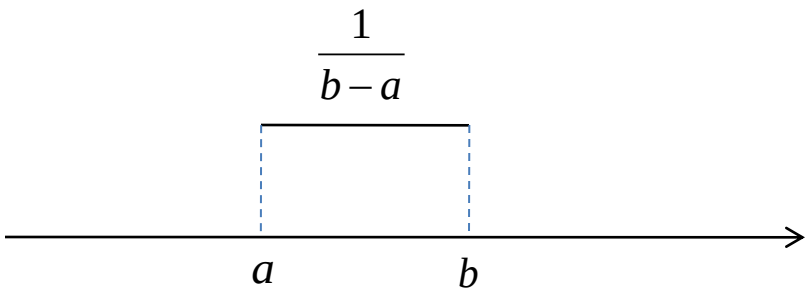
- **Example**
  - The pdf of a RV X has the form $f(x) = 5e^{-ax}u(x)$
    - The value of a
    - the probability that $X \leq 2$
    - The probability that $1 < X \leq 2$

CONTINUOUS RV

- **Uniform distribution**
  - A random variable X is said to be uniformly distributed over the interval (a, b) if its pdf is given by

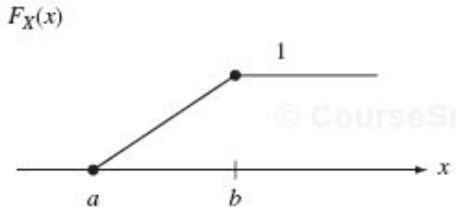
$$f_X(x) = \begin{cases} \dfrac{1}{b-a}, & a < x \le b \\ 0, & \text{otherwise} \end{cases}$$

  - The RV has equal probability of being any number inside (a, b)



CONTINUOUS RV

- **Uniform distribution**
  - CDF of a uniform RV



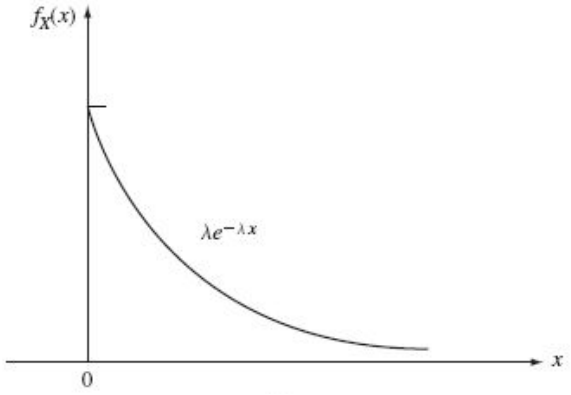
CONTINUOUS RV

- **Example**
  - A continuous RV X is uniformly distributed between (3, 8). Find the following probability
    - $\Pr(4 < X \leq 5)$
    - $\Pr(2 < X \leq 5)$
    - $\Pr(2 < X \leq 9)$

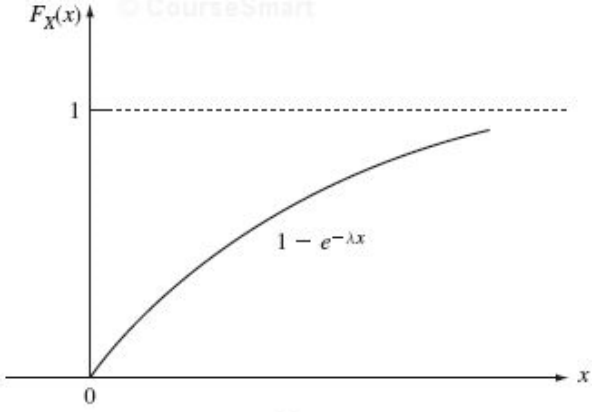
CONTINUOUS RV

- **Exponential RV**
  - The pdf of an exponential RV with parameter $\lambda > 0$

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$



  - The CDF of an exponential RV

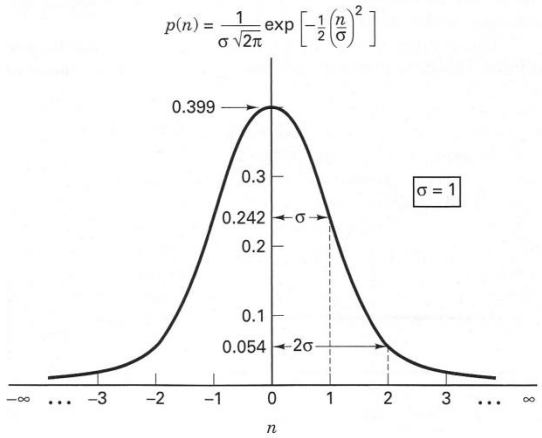


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CONTINUOUS RV

- **Normal RV (Gaussian RV)**
  - A Gaussian RV X with parameters m and σ^2

$$f(x) = \frac{1}{\sqrt{2\pi}\cdot\sigma}\exp\left[-\frac{(x-m)^2}{2\sigma^2}\right]$$





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MOMENTS: DISCRETE RV

- **The expected value, or mean, of a discrete RV, *X*, is**

$$E(X) = \sum_{x_i} x_i \, p_X(x_i)$$

  - The summation is performed over all the possible values of *X*.
  - Expected value, mean, average, first moment
  - Interpretation
    - The *weighted average* of the random variable
    - If the random variable takes all the values with equal probability, that is, $p_X(x_1) = p_X(x_2) = \cdots = p_X(x_N) = \frac{1}{N}$, then

$$E(X) = \frac{1}{N} \sum_{i=1}^{N} x_i$$

    - Given more weight to more probable values, and vice versa
  - Also denoted as

$$E(X) = \overline{X} = m_X$$

MOMENTS: DISCRETE RV

- **Example**
  - Consider an unfair coin with Pr(H) = 0.3. Define the RV X, such that X = 0 if H, and X = 1 if T. Find the mean of X. (Bernoulli RV)

  - Find the mean of a geometric RV with parameter *p*.

MOMENTS: DISCRETE RV

- **Example**
  - Find the mean of a binomial RV with parameters *n*, *k*, and *p*

MOMENTS: DISCRETE RV

- **Example**
  - Find the mean of the Poisson RV with the parameter λ

MOMENTS: DISCRETE RV

- **Expected value of any function of X**

$$E[g(X)] = \sum_{x_i} g(x_i)\, p_X(x_i)$$

  - $E(X)$: the expectation operator
  - The variable of the expectation operator must be a random variable
    - X is an RV
    - g(X) is an RV

  - Also denoted as $E[g(X)] = \overline{g(X)}$

MOMENTS: DISCRETE RV

- **Example**
  - Consider a gambling game of throwing 3 fair coins. Let X be the RV denoting the number of heads. The payoff is X^2, e.g. the Casino pays \$0 for 0 H, \$1 for 1H, \$4 for 2H, and \$9 for 3H. If the Casino charges \$3.5 per play, will you play the game?

MOMENTS: DISCRETE RV

- **Variance**
  - The variance of a discrete RV X is

$$\mathrm{Var}(X) = \sigma_X^2 = E\left[(X - m_X)^2\right] = \sum_{x_i} (x_i - m_X)^2 \, p_X(x_i)$$

  - Also called: the 2nd central moment of X
  - Standard deviation:

$$\sigma_X = \sqrt{\mathrm{Var}(X)} = \sqrt{\sum_{x_i} (x_i - m_X)^2}$$

  - Property

$$\sigma_X^2 = E\left(X^2\right) - m_X^2$$

$$\mathrm{Var}(X + c) =$$

  - Physical interpretation
    - How far away the random variable is from its expected value

MOMENTS: DISCRETE RV

- **Example**
  - Find the variance of the Bernoulli RV
  - Consider a deterministic variable x = 3. What is the mean and variance?

MOMENTS: DISCRETE RV

- **The *n*-th moment of the random variable X**

$$E[X^n] = \sum_{x_i} x_i^n \, p_X(x_i)$$

  - 1st moment: $E[X]$ mean
  - 2nd moment: $E[X^2]$

- **The n-th central moment of the random variable X**

$$E[(X - m_X)^n] = \sum_{x_i} (x_i - m_X)^n \, p_X(x_i)$$

  - 1st central moment: $E[X - m_X] =$
  - 2nd central moment: $E\left[(X - m_X)^2\right] = \sigma_X^2$ variance

- **The combination of all the moments, n = 1, 2, …., gives a complete description of the random variable**

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MOMENTS: DISCRETE RV

- **The expected value, or mean, of a continuous RV, *X*, is**

$$E(X) = \int_{-\infty}^{+\infty} x f_X(x) dx$$

  - Recall: discrete RV $E(X) = \sum_{x_i} x_i p_X(x_i)$
  - Expected value, mean, average, first moment
  - Interpretation
    - The *weighted average* of the random variable
    - Recall: integration is an extreme case of summation.
  - Also denoted as

$$E(X) = \overline{X} = m_X$$

MOMENTS: CONTINOUS RV

- **Example**
  - The mean of an RV X uniformly distributed in (a, b)

MOMENTS: CONTINOUS RV

- **Example**
  - The mean of an exponential RV with parameter λ

MOMENTS: CONTINUOUS RV

- **Expected value of any function of X**

$$E[g(X)] = \int_{-\infty}^{+\infty} g(x) f_X(x) dx$$

  - Recall: discrete RV $E[g(X)] = \sum_{x_i} g(x_i) p_X(x_i)$
  - $E(X)$: the expectation operator
  - The variable of the expectation operator must be a random variable
    - X is an RV
    - g(X) is an RV
  - Also denoted as

$$E[g(X)] = \overline{g(X)}$$

MOMENTS: CONTINUOUS RV

- **Variance**
  - The variance of a continuous RV X is

$$\mathrm{Var}(X) = \sigma_X^2 = E\left[(X - m_X)^2\right] = \int_{-\infty}^{+\infty} (x - m_X)^2 \, f_X(x) dx$$

  - Recall: discrete RV $\sigma_X^2 = \sum_{x_i} (x_i - m_X)^2 \, p_X(x_i)$
  - Also called: the 2nd central moment of X
  - Standard deviation: σ_X
  - Property

$$\sigma_X^2 = E\left(X^2\right) - m_X^2$$

$$\mathrm{Var}(X + c) =$$

  - Physical interpretation
    - How far away the random variable is from its expected value

MOMENTS: DISCRETE RV

- **The *n*-th moment of the random variable X**

$$E[X^n] = \int_{-\infty}^{+\infty} x^n f_X(x)dx$$

  - 1st moment: $E[X]$ mean
  - 2nd moment: $E[X^2]$

- **The n-th central moment of the random variable X**

$$E[(X - m_X)^n] = \int_{-\infty}^{+\infty} (x - m_X)^n f_X(x)dx$$

  - 1st central moment: $E[X - m_X] =$
  - 2nd central moment: $E\left[(X - m_X)^2\right] = \sigma_X^2$ variance

- **The combination of all the moments, n = 1, 2, …., gives a complete description of the random variable**

MOMENTS: CONTINUOUS RV

- **Example**
  - Find the variance of an RV X uniformly distributed on (a, b)

MOMENTS

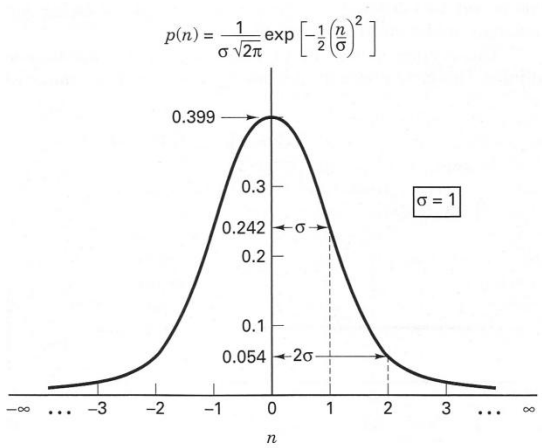
- **Example**
  - Consider a random variable $Y = A\cos(2\pi f + \Theta)$, where A, f are constants, and Θ is uniformly distributed in $[0,2\pi]$. Find the expected value of *Y*.

CONTINUOUS RV

- **Normal RV (Gaussian RV)**

$$f(x) = \frac{1}{\sqrt{2\pi}\cdot\sigma}\exp\left[-\frac{(x-m)^2}{2\sigma^2}\right]$$

  - Find the mean and variance



  - The Gaussian RV is completely determined by its first 2 moments.
    - No true for other RVs.

MOMENTS: CONTINUOUS RV

- **Gaussian RV**
  - Define: Q function

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty \exp\left(-\frac{u^2}{2}\right) du$$

  - For a Gaussian RV X with mean *m* and variance σ^2 . What is the probability that X < m - σ

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FUNCTIONS OF ONE RV

- **Example**
  - X is an RV with pdf $f_X(x)$. Let Y = a X + b, where a and b are constants. What is the pdf of Y?

  - If X is a Gaussian RV with mean 0 and variance 1. Find the pdf of

    $Y = \sigma X + m$

FUNCTIONS OF ONE RV

- **Example**
  - If X is a Gaussian RV with mean 0 and variance 1. Find the pdf of $Y = X^2$

FUNCTIONS OF ONE RV

- **Example**
  - Let X be an exponential RV with parameter $\lambda = 2$. Find the pdf of $Y = X^2$